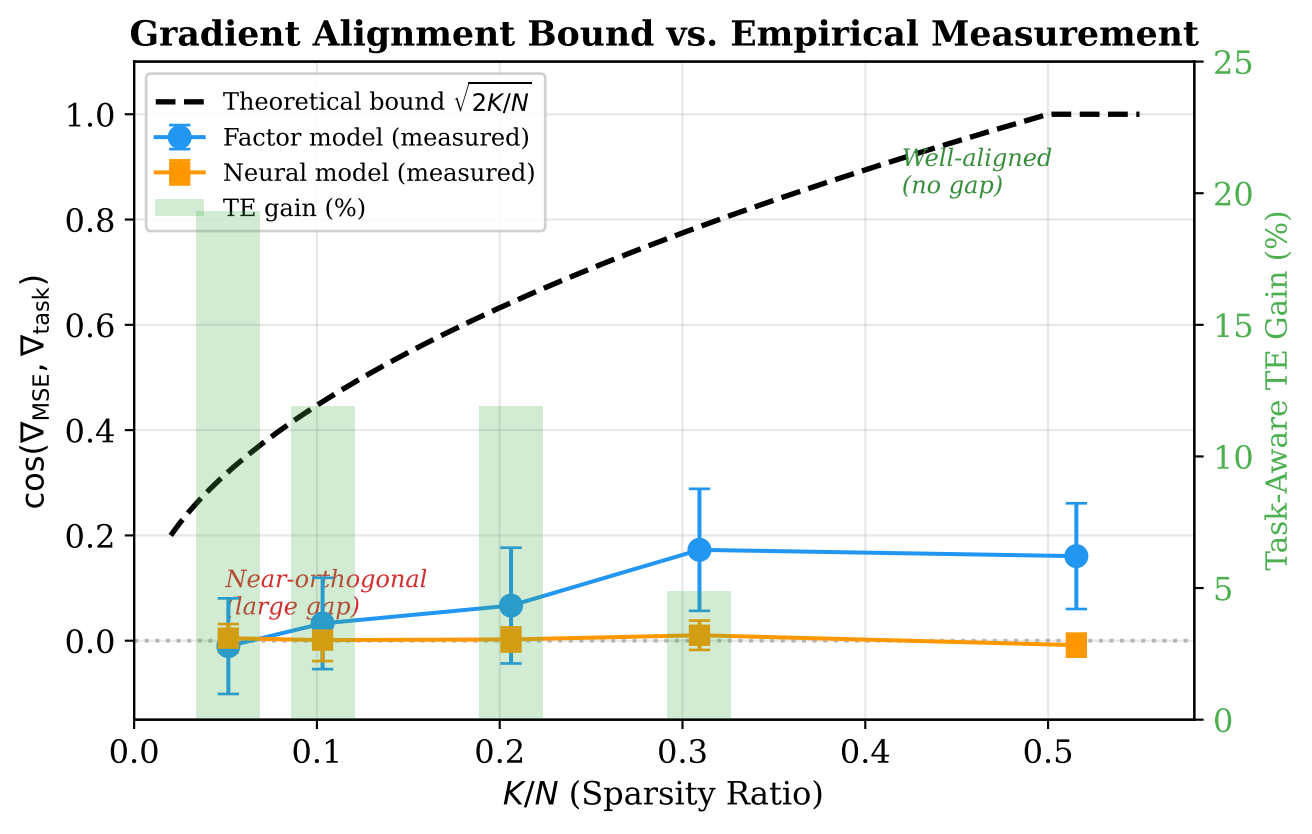
Gradient Alignment Bound vs. Empirical Measurement
Theoretical bound $\sqrt{2K/N}$
Factor model (measured)
Neural model (measured)
TE gain (%)
Well-aligned
(no gap)
Near-orthogonal
(large gap)
$\cos(\nabla_{MSE}, \nabla_{task})$
Task-Aware TE Gain (%)
K/N (Sparsity Ratio)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
0.1
0.3
0.5